Living Mobility Datasets from Public Video: A Continuously Running Pipeline for Crashes, Near-Collisions and Pedestrian Environments

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***Abstract*— Automated vehicles need data on crashes and on where people walk, but research driving datasets contain few crashes and come from few cities. Dashcam and walking videos on YouTube show both, but they must be screened, cut into segments and labelled before use. We present a pipeline that does this without manual work and keeps running, so the datasets grow as new videos are uploaded. A language model screens each video's title and description, a video vision-language model (VLM) checks the cuts and labels each segment, and a location is recorded only when the video states it. Every decision is stored with its evidence, so each segment is traceable to its source video. We apply the pipeline to road traffic crashes and to walking. The snapshot of 9 November 2026 holds 74,315 crash and near-collision segments (295.9 h) from 2,848 videos in 177 countries, and 63,993 walking segments (3,088.9 h) from 3,496 videos in 103 countries. In a validation by two independent reviewers, the model's camera view, time of day and weather agreed with them in 94–97 % of crash clips, its walking decisions in all walking clips, and the recorded locality matched the footage in 82 of 87 clips.**

I. INTRODUCTION

Road traffic crashes killed 1.19 million people in 2021, and 53% of them were pedestrians, cyclists, motorcyclists or users of micromobility [1]. To develop and test automated vehicles for this reality, researchers need two kinds of data that are hard to record on purpose: rare safety-critical events, and the many places where vulnerable road users move. Driving datasets such as KITTI, Cityscapes, BDD100K, nuScenes and the Waymo Open Dataset [2]–[6], and pedestrian datasets such as JAAD and PIE [7], [8], have calibrated sensors and careful labels, but few crashes and few cities. Crash video datasets [9]–[14] do contain crashes, but each was collected once, holds at most 4,677 videos [13] and comes from a few regions.

Public video platforms fill both gaps. Dashcam owners upload collisions, near-misses and long drives through cities worldwide [15], and walking-tour channels publish hours of footage filmed on foot [16]. There is a vast amount of such video [17], but it is not ready for research. Search results include games, simulations and commentary. Compilations join events from different places. Long videos start with intros. The location is often missing or wrong, and videos are removed. To turn this material into data, it has to be screened, split in time and labelled, with a record kept of every step.

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Vision-language models (VLMs) can now review video segment by segment at this scale [18]–[20], but their answers are not ground truth. We use a model only where the evidence is in the picture, store every decision with its evidence, and check the resulting labels both automatically and by hand.

Existing dashcam crash datasets were each built for one task: anticipating accidents [9], [11], driver attention [12], anomaly detection [10], [13] or predicting collisions [14]. They are labelled by hand and released once as fixed benchmarks. Our crash corpus has 74,315 segments from 2,848 videos, is placed on the map where the evidence allows, and keeps growing. For pedestrians, JAAD and PIE label behaviour seen from a vehicle in a few cities [7], [8], and walking-tour videos are known to hold rich street-level information [16]. Our walking corpus shows the pedestrian's own view in 1,496 localities. Datasheets [21] ask how a dataset was collected; for datasets built by models, this includes which model produced each label and on what evidence.

A. Aim of the Study

This study shows that a pipeline running without supervision can turn public video into mobility datasets that keep growing, with every record traceable to its source video and to the decisions that produced it. Two use cases test different parts of the pipeline: short safety-critical events and long, continuous walking footage. The study makes four contributions: a pipeline that can stop and resume, keeps versions and stores the evidence for every decision, with one shared core and two domain branches (Sec. II); two growing corpora, currently 74,315 crash and 63,993 walking segments from 177 and 103 countries, with their coverage compared with where road deaths occur (Sec. III); evidence that the collection is *living*, from 69 days of growth without manual work (Fig. 3); and an automatic consistency check and a two-reviewer validation that show which model labels can be used (Sec. III-C).

II. METHOD

Fig. 1 shows the stages. Each stage writes its decision, its confidence and its evidence into a stored record for each video. This has three benefits. First, collection can stop at any point and resume without repeating finished work. Second, an improved stage can be re-run on only the records that need it, leaving the rest unchanged. Third, every released record can be traced to its source video, the screening decision, its start and end times, and the model version that labelled it. Cheap checks come before expensive ones: the uploader's

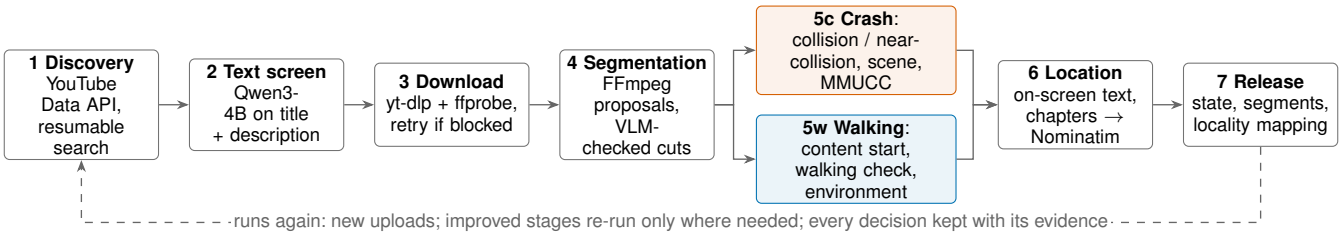


Fig. 1. The pipeline. Stages 1–4, 6 and 7 are shared by both corpora; stage 5 accepts and labels segments for each domain. Because every decision is stored, any stage can be re-run without repeating the others.

text is screened before any video is downloaded, and a VLM is used only for decisions that need the picture.

A. Collecting Videos

Candidate videos are found through the YouTube Data API with a fixed set of search queries for each corpus (Table I). English queries favour English-speaking uploaders, so the crash corpus also uses one query in each of 14 other languages. The pipeline stores how far it has paged through each query, so it resumes where it stopped and never adds the same video twice. Before download, Qwen3-4B-Instruct-2507 [22] reads the title and the start of the description and judges whether the video is likely to contain the footage we want. This is a quick plausibility check, not the decision to include a video: it avoids analysing videos whose own description gives no reason to expect relevant footage. Videos that pass are downloaded with yt-dlp and checked with ffprobe. When YouTube blocks a download with a bot check, the video is kept for a later attempt rather than rejected; in the current snapshot this applies to 2,017 crash candidates.

FFmpeg's scene-change detection proposes cuts, but camera shake, fast turns and changes in exposure can look like edits. Cosmos3-Nano [18] therefore looks at the 6 s around each proposed cut (3 s on either side) and keeps only real cuts. Compilations are split into one segment per event, and walking videos into continuous stretches.

TABLE I
SEARCH AND TEXT-SCREENING SETTINGS.

	Crash	Walking
Queries	'car crash compilation dashcam', 'traffic accident compilation dashcam', 'road crash compilation CCTV', 'car accidents caught on camera', 'accidentes de tránsito captados en cámara', 'acidentes de trânsito flagrados por câmeras', 'ДТП видеорегистратор подборка', 'सड़क दुर्घटना कैमरे में कैद', 'trafik kazası kamera kaydı', 'مراقبة كاميرا سيارات حوادث', 'kecelakaan lalu lintas terekam kamera', '行车记录仪 交通事故', 'ドライブレコーダー 事故', '블랙박스 사고 영상', 'accident de la route caméra embarquée', 'tai nạn giao thông camera hành trình', 'wypadki samochodowe kamera', 'Unfall Dashcam Deutschland'	'walking tour', 'city walk', 'walk with me', 'virtual walk', '4k walking tour', 'POV walking', 'street walk', 'nature walk', 'rain walk', 'night walk' (all in English)
Pages per query	4 × 50 results	2 × 50 results
Duration	no limit	≥300 s
Text read	title + 5,000 characters of description	title + 4,000 characters of description
Threshold	0.65	0.60
Excluded	games, simulations, films, crash tests, motorsport, toy vehicles, commentary without footage, aftermath only	workouts, treadmills, games, animation, music videos, reviews, driving or cycling, public transport rides, drone footage, flights, talking heads, slideshows

B. Labelling Segments

Crash branch. For each candidate segment, the VLM sees frames from its whole length and decides whether it shows a real road traffic collision, or a near-collision in which someone avoids the impact at the last moment. It rejects games, simulations, staged scenes, crash tests, motorsport, commentary and footage that shows only the aftermath. The model also returns its confidence, a description, the camera view, the road users, the road environment, the time of day, the weather, the road surface, the visible outcomes and, if visible, the time of impact. Collisions are further described with terms from the sixth edition of the Model Minimum Uniform Crash Criteria (MMUCC) of the US National Highway Traffic Safety Administration (NHTSA) [23]: the first harmful event, the manner of collision when another motor vehicle is involved, and up to four events in sequence. The model is told not to judge fault, intent, impairment or injury severity. Each answer is checked against a fixed format and for internal consistency, and the question is asked again if the answer is invalid.

The MMUCC fields are labelled in a separate taxonomy stage, which works in two steps. First, it asks when the contact happens, showing the whole segment with the playback time printed on each frame. Second, it takes up to three windows of ±2 s, around this estimate and around the two largest changes between frames, and samples them at 10 frames per second. On the earliest window that shows contact, the model answers plain questions: what hit what, which parts of the vehicles touched, and which way each was moving. Fixed rules turn these answers into the MMUCC manner of collision and first harmful event.

Walking branch. The VLM first finds where the walking footage starts, skipping intros and title cards. The rest is divided into windows, each checked at its start, middle and end. A window is kept only if the viewpoint and motion show a person walking through a real place, not footage from a drone, vehicle or bicycle, a static camera or a talking head.

TABLE II
CORPUS SNAPSHOT OF 9 NOVEMBER 2026.

	Crash	Walking
Candidate videos found	8,125	7,725
passed text screening	5,000	6,436
accepted on the footage	2,848	3,496
Accepted segments	74,315	63,993
Footage (h)	295.9	3,088.9
Median segment length (s)	10.4	215
Collisions / near-collisions	62,854 / 5,939[a]	–
Night / dawn or dusk segments (%)	13.1 / 4.1	33.9 / 9.4
Segments placed in a locality	25,228 (33.9%)	53,428 (83.5%)
Localities / countries	7,204 / 177	1,496 / 103

[a] A further 5,522 segments are waiting for the labelling stage to be re-run.

Each kept window is given a time of day and one of twelve walking environments.

C. Locating Segments

Guessing a place from scenery or language is unreliable, so we do not do it. A location is recorded only from explicit evidence: text readable in the footage, such as overlays, signs or GPS stamps (when the VLM's confidence is at least 0.90), the uploader's chapters and timestamps, and the video's metadata. Nominatim [24] resolves each candidate to an inhabited locality, and we publish the locality's reference point, not the camera's position. Segments without enough evidence stay in the corpus, with the reason recorded.

III. RESULTS

A. Size, Origin and Growth

Table II summarises the current snapshot; all counts grow with every run. Of 8,125 crash candidates, 3,125 were rejected on their text, 2,089 could not yet be downloaded (2,017 because of a bot check, 69 age-restricted, 3 removed), 63 were rejected on the footage, and 2,848 were accepted, giving 74,315 segments. Of 7,725 walking candidates, 1,289 were rejected on their text, 2,939 on the footage and one at cut checking, leaving 3,496 videos and 63,993 segments. The two corpora lose candidates at different stages (Fig. 3b). Crash queries attract uploads whose descriptions already reveal games or commentary. Walking queries return drone flights, vehicle rides or talking heads with plausible descriptions, which only the visual check can reject.

Fig. 2 shows the localities. Of the 25,228 placed crash segments, 16,042 (63.6%) are in North America, followed by Europe (4,756), Asia (2,005), Oceania (1,064), Africa (707) and South America (613). The United States alone has 14,516, because of its large dashcam channels. Walking segments are split between Europe (17,607), North America (17,599) and Asia (15,655), with fewer in Africa (1,014), South America (817) and Oceania (732). The most frequent countries are the United States (16,009), the United Kingdom (5,390), Japan (2,918) and Italy (2,372). A locality is known for 83.5% of walking segments but only 33.9% of crash segments: walking uploads usually name their city, whereas crash compilations combine clips from many places and rarely say where.

We compared each region's share of placed videos with its share of the world's road deaths in 2021 [1]. The ratio of the two shares is a representation index: 1 means a region appears as often as its road deaths would suggest. The 1,059 crash videos with a known country come from 127 countries that together have 93.1% of road deaths, yet five countries hold 49.7% of these videos. Asia and Africa have 83.6% of road deaths but only 24.4% of the videos (index 0.28 and 0.31), while Europe (7.6) and North America (5.0) are over-represented. Walking videos cover Asia better (0.59) but Africa less (0.11).

Fig. 3a shows the released corpora from 1 September to 9 November 2026 (69 days), in which the pipeline ran without manual work. The crash corpus grew from 34,295 segments from 902 videos on 10 September to 74,315 segments from 2,848 videos on 9 November, and from 151 to 296 hours of footage. The walking corpus grew from 3,820 segments (146 videos, 198 h) on 1 September to 63,993 segments (3,496 videos, 3,089 h). Each snapshot is a commit of the released files in the public repository, so a study can cite the exact version it used while the collection keeps growing.

B. Content of the Corpora

The model labels 62,854 of the 74,315 crash segments as collisions and 5,939 as near-collisions (Fig. 4); Sec. III-C reports how often reviewers agree. Dashcams filmed 94.7% of the segments and closed-circuit television (CCTV) 2.8%; the rest come from action cameras, broadcasts and handheld cameras. Motorways account for 51.5% of segments, urban roads for 16.3%, rural roads for 10.6%, junctions for 7.6%, car parks for 5.6% and intersections for 2.6%. These shares are model labels, which agreed with the reviewers' consensus for 64.9% of clips (Table III). The corpus has many conditions that are rare in planned recordings: 13.1% of segments are at night, 4.5% on snow or ice, 3.0% on wet roads and 1.9% in rain. Vulnerable road users appear too: pedestrians in 2,307 segments, motorcycles in 3,864 and cyclists in 881 (Table IV). The model describes the road environment in its own words, giving 176 distinct values such as 'toll plaza' or 'level crossing'; Fig. 4 groups them into eight classes.

Streets make up 61.7% of walking segments, followed by mixed environments (12.0%), markets (7.0%), parks and nature (6.5%), indoor spaces (5.2%), squares (2.9%), waterfronts (2.5%) and trails (1.7%). Night walks are a popular genre, so 33.9% of walking segments are at night and 9.4% at dawn or dusk, far more low-light footage than in the crash corpus (Fig. 4c). The mix differs between regions (Fig. 5): markets make up 12.0% of placed segments in Asia, 16.4% in South America and 23.4% in Africa, but 1.5% in North America, where 78.7% are streets. For 35,373 segments (55.3%), the location evidence is specific to the segment: text in the footage supports 17,904 segments, the uploader's timestamps or chapters 11,678, and both 5,791. The other 28,620 segments rely only on metadata for the whole video.

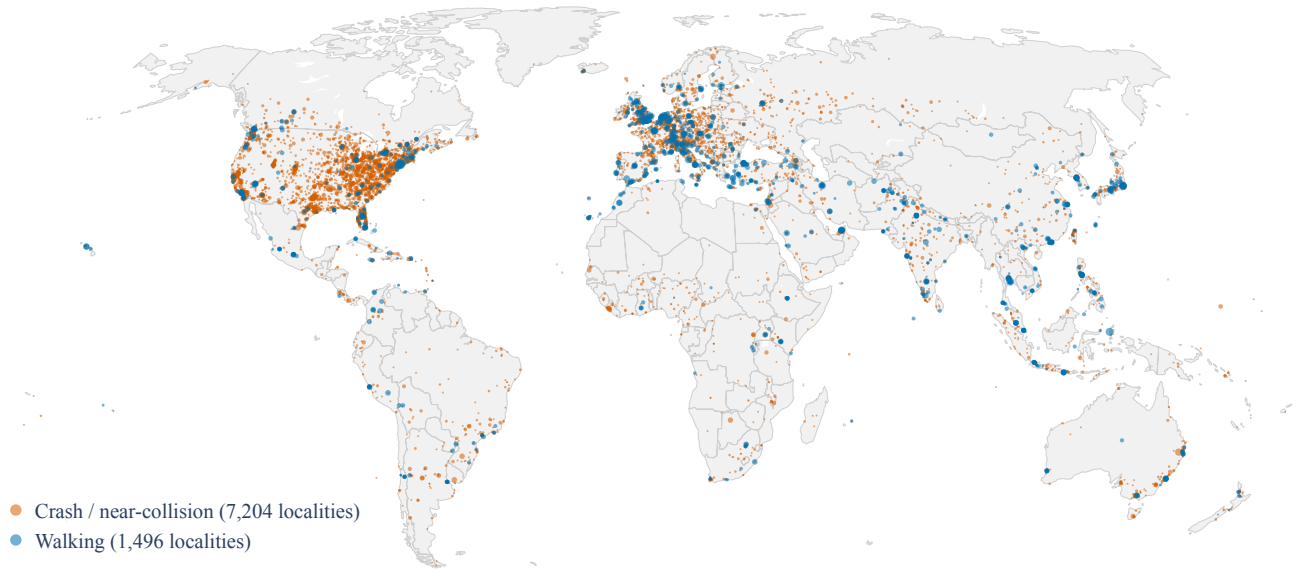
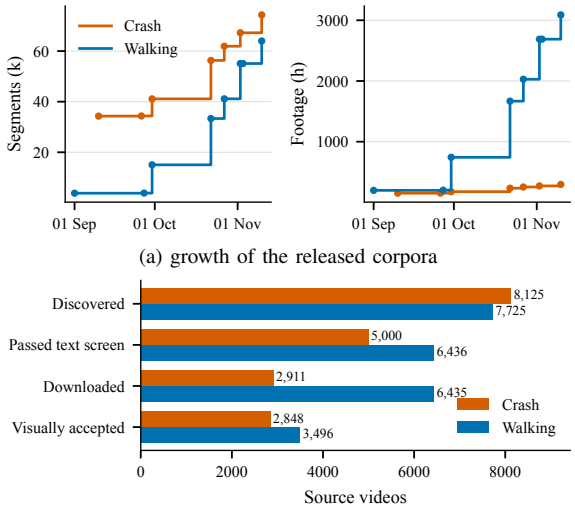


Fig. 2. Localities with at least one placed segment. Marker size grows with the number of segments (log scale). Crash: 25,228 segments; walking: 53,428 segments.



(a) growth of the released corpora

(b) source videos remaining after each stage

Fig. 3. The collection is *living*. (a) Released segments and footage at each published snapshot from 1 September to 9 November 2026, without manual work. (b) The stages at which candidates leave the pipeline.

TABLE III

MANUAL VALIDATION: AGREEMENT BETWEEN THE TWO REVIEWERS (%, COHEN'S κ) AND BETWEEN THE MODEL AND THE REVIEWERS' CONSENSUS (MODEL LABELS AS STORED WHEN THE SAMPLE WAS DRAWN; MANNER OF COLLISION FROM THE TAXONOMY STAGE).

	Reviewer vs reviewer			Model vs consensus	
Label	n	%	κ	n	%
Crash					
Event kind[a]	90	65.6	0.50	59	54.2
Manner of collision	36	69.4	0.62	20	30.0
Camera view	75	94.7	0.71	71	94.4
Time of day	75	90.7	0.66	68	97.1
Weather	75	86.7	0.15	65	95.4
Road environment	75	49.3	0.30	37	64.9
Walking					
Walking content	99	99.0	–	98	100.0
Time of day[b]	98	83.7	0.70	78	98.7
Walking environment	98	57.1	0.31	56	80.4

[a] Collision, near-collision, no event, not real traffic, or unusable. [b] Consensus answers of 'can't tell' left out.

C. Validation of the Labels

Checking labels by hand is the definitive test, but it cannot be repeated for every snapshot of a living collection. We therefore combine two checks. An automatic audit runs on every snapshot without running any model: it compares fields that the model produced separately for the same segment and flags combinations that cannot both be true. A manual validation checks a stratified sample.

Two of the authors reviewed a sample drawn on 5 November. It was stratified by continent (and, for crashes, by event kind) and held at most one segment per video. Clips were played from YouTube in a review page. The reviewers answered the same multiple-choice questions as the model, with 'can't tell' and 'unusable' options. They labelled independently, without seeing each other's answers or the model's. Before starting, they read a reviewer guide (supplementary material) that defines every answer with tie-break rules and example clips not in the sample, and covers the different kinds of crash, including a diagram of each manner of collision. Both labelled the same 100 crash and 100 walking clips, of which 90 and 99 could be played; the other half of the sample is still to be reviewed.

Where the reviewers agree, the model matches them on camera view, time of day and weather in 94–97% of crash clips, and on walking content in all walking clips (Table III).

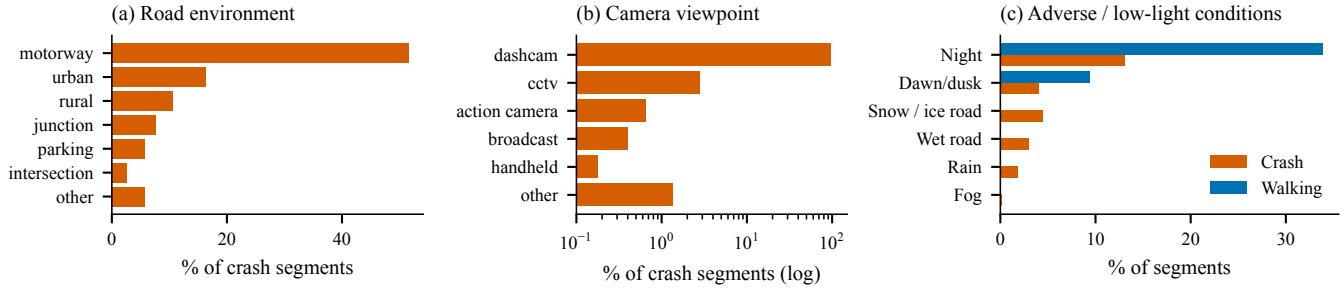


Fig. 4. Scenes in the crash corpus. (a) Road environment and (b) camera view (log scale) as a share of crash segments; (c) low-light and adverse conditions in the crash and walking corpora.

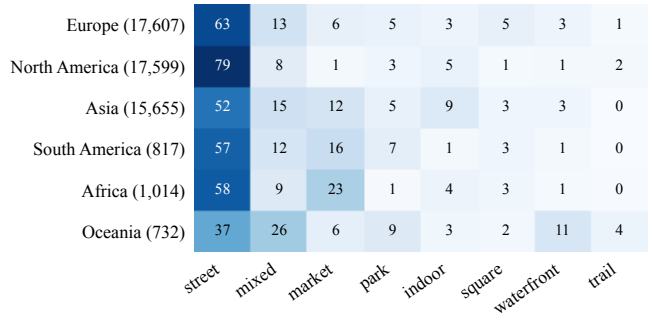


Fig. 5. Walking environment by continent (% of placed walking segments in each row; segment counts in brackets).

Crash acceptance is weaker. Of the 59 crash clips with a consensus, 44 (75%) show a real collision or near-collision, 10 show no event and 5 show no real traffic. The model tells collisions from near-collisions correctly for 32 of the 44. Road and walking environments are hard even for people (49% and 57% agreement between reviewers). The locality matched the footage for 20 of 24 crash clips and 62 of 63 walking clips; all three mismatches were crash clips. In 10 of the 15 false crash clips the reviewers saw ordinary driving with no event. In the other 5 they saw no real traffic; their notes mention a still photograph, a banner shown before the footage and an interview.

We tested the taxonomy stage on the clips on which both reviewers agreed (36 with an agreed event, 20 with an agreed manner of collision). It finds the moment of contact in 18 of 30 collisions, tells collisions from near-collisions in 21 of 36, and matches the manner of collision in 6 of 20, against 69% agreement between the reviewers. Qwen3-VL-32B [22] used in the same way matched 7 of 20, so model size is not the only limit. Because these clips were also used to tune the stage, the figures are optimistic. The MMUCC fields are released with their measured agreement and are meant for finding candidate clips to confirm by hand.

D. Searching for Scenarios

The structured labels make the corpora searchable. The cases in Table IV have between 193 and 3,864 segments each, from between 16 and 84 countries.

TABLE IV

EXAMPLE SCENARIO SEARCHES IN THE CRASH CORPUS (STRUCTURED FIELDS ONLY; COUNTRIES COUNTED OVER PLACED SEGMENTS).

Scenario	Segments	Videos	Countries
Pedestrian involved	2,307	861	67
at night	351	207	25
Cyclist involved	881	383	52
Motorcycle involved	3,864	898	83
Bus or lorry, urban road	2,245	845	84
Near-collision at an intersection or junction	689	384	50
Collision on snow or ice	2,991	501	78
Night, wet road	796	414	42
Roundabout	265	181	31
Railway level crossing	193	85	16

IV. DISCUSSION

The corpora can be used for (i) *finding scenarios* for testing in simulation; (ii) *training and testing accident anticipation and detection* beyond existing benchmarks [9], [11]–[13], with splits by country to test how well models generalise to new places; (iii) *studying pedestrian environments* for the perception of vulnerable road users; and (iv) *evaluating VLMs* on the same inputs, since each segment stores the model's answers and evidence.

The pipeline is not tied to these two use cases. The shared core, which searches, screens, downloads, cuts and locates, stays the same; a new corpus needs only its own search queries, the question used to screen titles and descriptions, and a domain branch that labels its segments. Corpora of cycling, e-scooters or other traffic situations could be built in this way and would grow in the same way as new videos are uploaded.

The validation also shows how the labels should be used. Camera view, time of day, weather and walking acceptance can be used as they are, for example to filter footage or to build splits. Crash acceptance gives a list of candidates to confirm before use. Road and walking environments are hard even for people, so they suit broad filtering rather than detailed analysis. The manner of collision is best used to rank clips for a person to confirm. A location identifies a locality, which suits comparisons between regions but not the study of a particular junction.

Public video can therefore feed datasets of safety-critical driving events and pedestrian environments that are living, global and traceable. One pipeline produced 74,315 crash

and 63,993 walking segments from 177 and 103 countries, and in 69 days without manual work the crash corpus grew from 34,295 and the walking corpus from 3,820 segments, although Asia and Africa remain under-represented. The validation showed that scene labels, walking acceptance and locations are reliable, and showed which other labels should be confirmed by hand, a check that every dataset built with VLMs needs.

V. LIMITATIONS AND FUTURE WORK

Public video is not a representative sample of mobility. What it covers depends on who records and uploads, on language, tourism, camera ownership and internet access, on the platform's ranking, and on our queries. The United States dominates the crash corpus, and the counts say nothing about crash risk or exposure. The labels are automatic and were validated on one sample of 189 segments by two of the authors. In 15 of 59 reviewed crash segments (25.4%) there was no real crash, so crash counts overstate the number of usable events, and the manner of collision is not reliable for individual segments. Videos can be removed after release, and public footage may show identifiable people. We therefore release no video files, only identifiers, time intervals and derived labels, and we remove records on request. Finally, the pipeline depends on models and services that change, so each record stores the versions that produced it (e.g. `cosmos3_full_clip_crash_v3` and taxonomy `nhtsa_mmucc6_video_v3`).

Future work follows from these limits. We will make crash acceptance stricter with a second, focused question on each accepted segment, so that fewer segments without a real crash enter the corpus, and review the second half of the validation sample, which was not used to tune the taxonomy stage. The crash corpus already searches in 14 languages besides English, which should improve its coverage of Asia and Africa; we will track this against road deaths in every snapshot. The walking corpus still searches only in English, and adding queries in the same languages is a simple way to improve its coverage, especially of Africa. Finally, we will locate segments from individual frames, to place more of the crash corpus.

VI. SUPPLEMENTARY MATERIAL

The pipelines and the released corpora are available at `https://github.com/crowd-dataset/car-crash-yt` and `https://github.com/crowd-dataset/walking-yt`. The released files contain YouTube identifiers, time intervals, labels and records of each decision. The crash repository also holds the validation kit, including the reviewer guide (`validation/GUIDE.md`), and the coverage analysis.

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